

Lectures On Diffeology



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These lecture notes are intended for students interested in differential geometry, particularly in situations not covered by the classical theory. The first part is a series of lectures given at Shantou University as a special program. They introduce the main areas of differential geometry extended to diffeology, as developed in the chapters of the diffeology textbook. The second part consists of a series of notes and exercises chosen because they do not fall within the scope of the theory of manifolds. They illustrate some applications of diffeology: infinite spaces with singular quotients, symplectic diffeological space without Hamiltonian diffeomorphisms, general spaces of geodesics, Riemannian diffeology program, classifying space of quasi-spheres, and other examples. They have been chosen to familiarise the student with some specific techniques used in the versatile environment of diffeology.

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